Pay for Training, Not for Openness: Reward Design for Original and Distilled AI Models

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FINAL · CAPABILITY-TRANSFER OPENNESS

KEY MESSAGE

Private and social rankings are exact reverses: the developer's best move is society's worst. So markets rest on secrecy, or fill with open copies, unless the prize is paid per unit of **training compute**, not for **openness** alone.

01 Question & model

Research question & verified gap

Q A developer may imitate (distil) or innovate (train from scratch), and may release the result under secrecy or openness. Which of the four does it choose — and does society want the same?

Gap Rules that reward openness reward it as such (Qiu et al. 2025), and stay silent on whether a model was originally trained or merely imitated. That silence is where the incentive breaks.

Why it matters A prize that pays for openness alone pays the imitator, not the innovator.

Strategic model & foundations

Players N firms each choose sourcing (imitate or innovate) and release (secrecy or openness).

Payoffs four cells (Figure 1): secrecy keeps the monopoly profit; a release earns a prize on its compute and loses the edge.

Solution concept Nash equilibrium, set against society's ranking.

Assumptions innovation costs more and uses more compute than imitation; the prize rate k is paid per unit of compute E.

Notation C_d, E_d = cost and compute of imitation · C_n, E_n = cost and compute of innovation · k = prize per unit of E · L = loss from disclosure · π = monopoly profit kept under secrecy · $\beta\phi K \cdot m/(m+m_0)$ = future share an open release buys (β discount · ϕ ecosystem efficiency · K external contributions · m compatibility share · m_0 internal scale).

Three complementary lenses

Game theory left alone, the best response is secrecy plus imitation — the cheap way to keep the monopoly price.

Social choice society ranks openness plus innovation best and secrecy plus imitation worst; the two orders are exact reverses.

Mechanism design a prize paid for openness alone rewards the imitator more than the innovator, who paid more to get there; paid per unit of compute it separates them.

Implemented payment rule & auction analogy

Implemented rule: proportional kE subsidy — each open release is paid k per unit of audited training compute E.
Flat and capped rewards are counterfactual designs, not this institution.
All-pay reading (Baye et al. 1996) is an analogy only: every firm pays its sunk training cost.

reward rule	(a) no motive, βϕK = 0	(b) motive on, βϕK = 19.98
none, k = 0	.96/.00/.04/.00 ①	.28/.68/.01/.03 ②
openness, R = 20	.11/.85/.00/.04 ②	.01/.95/.00/.04 ②
indexed, k = 4	.35/.00/.01/.64 ③	.03/.02/.00/.95 ③
indexed, capped, k = 12	.42/.54/.02/.03 ②	.03/.93/.00/.05 ②

Acknowledgments & contributions

Thanks to Prof. Luyao Zhang; Prof. Kenneth S. (Ken) Rogerson (Sept 7, 2026 workshop discussant); Han Zhang; reviewers Mustafa Ayub Khan, Jiayang Sun, Temur Akhtamjonov, Xuantong Fu, Feiyu Li; Tencent's Shanghai office; the Shanghai Science and Technology Museum; and the COMSCI/ECON 206 community.
Zhengjun He: compute-indexed reward idea, market-competition model, simulation and verification. Yichen Shen: three-question framing, Hugging Face Space, paper/poster consistency. AI assistance disclosed in the paper.

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02 Design & evidence

Figure 1 — the four strategy combinations

	Imitate · cheap (C_d, E_d)	Innovate · costly (C_n, E_n)
Secrecy	① Secrecy + imitation **Free-rider** — takes, gives nothing $\pi - C_d$	③ Secrecy + innovation **Monopolist** — builds, keeps it $\pi - C_n$
Openness	② Openness + imitation **Passer-on** — takes and gives back $-C_d + kE_d - L + \beta\phi K \cdot m/(m+m_0)$	④ Openness + innovation **Source** — builds, and shares $-C_n + kE_n - L + \beta\phi K \cdot m/(m+m_0)$

Figure 1. The strategy space is a 2 × 2 — sourcing (imitate or innovate) × release (secrecy or openness) — with each cell's payoff. Secrecy plus imitation keeps the monopoly price cheaply, which is why it dominates privately; openness plus innovation is the only cell that creates a reusable source, and it is the most expensive. Cells recomputed in **verify_k.py**. An open release also keeps the future market share it buys, $\beta\phi K \cdot m/(m+m_0)$ — the discounted value of the external contributions an open model attracts.

Figure 2 — private desire vs. social desirability

Combination	Private desire	Social desirability	Conflict
① Secrecy + imitation	■■■■	■□□□	← Largest conflict
③ Secrecy + innovation	■■■□	■■□□	← Small conflict
② Openness + imitation	■■□□	■■■□	← Small conflict
④ Openness + innovation	■□□□	■■■■	← Largest conflict

Dark red = private desire · green = social desirability (more squares = more wanted).

Three market states — the prize rate k decides which cell wins

Market state	Prize rate	Composition of the market	Outcome
State ① all secrecy	too small for disclosure	① dominates if βϕK is small	monopoly price; no source
State ② all imitation	disclosure, not innovation	② dominates — nobody innovates	no source is ever built
State ③ healthy	pays for both	④ wins; some secrecy survives	fair price and a live source

Figure 2. The private order (①> ③> ②> ④) and the social order (④> ②> ③> ①) are exact mirror images: the developer's favourite is society's least wanted. The private order holds while $\pi + L - \beta\phi K > C_n - C_d$, i.e. while the future-share motive is weak; with a strong motive openness wins even at k = 0 (panel b).

Interpretation. A prize too small buys no disclosure; one that pays for openness but not for cost buys copies, not capability; only a prize covering both keeps a live source.

Verification. Cells and states are recomputed from scratch in the checker, independently of the symbolic derivation; an interactive page lets a reader move the parameters and watch the winning cell flip.

Runs. Paying for openness alone fills the market with imitation and dries up the sources; paying per unit of compute turns it back to original release (sources: 0.04 → 0.95); capping the prize undoes that, and with the future-share motive opening pays only for m ∈ (0.101, 1.519). Fresh run: seed 206. Copying a copy adds no capability, so mutual imitation burns compute while capability decays — yet some imitation is still efficient, because it stops every firm from re-deriving the same result. Parameters are illustrative, not estimated.

Selected references

[1] Qiu, T., Laufer, B., Kleinberg, J. & Heidari, H. (2025). Modeling the Economic Impacts of AI Openness Regulation. NeurIPS 2025. arXiv:2507.14193 · doi:10.52202/085713-1246
[2] Xu, F., Wang, X., Chen, W. & Xie, K. (2025). The Economics of AI Foundation Models. arXiv:2510.15200 · doi:10.2139/ssrn.4999355
[3] Habibi, M. (2025). Open Sourcing GPTs. arXiv:2501.11581

03 Merits · Impacts · Roadmap

Intellectual Merits — three closest papers

Qiu et al. (2025) — openness regulation; we inherit openness as a strategic choice but separate original training from distillation.
Xu et al. (2025) — openness, competition, governance; we inherit the subsidy framing but index the reward to compute.
Habibi (2025) — why firms open-source; we inherit the future-share F(m) but compare flat vs. compute-indexed rewards.
Distinct: the four-decision structure, the compute index, the three market states. Evidence: analytical + simulation. Model-based findings, not real-firm effects.

Practical Impacts — SDG 9.5 · 10.2

Mechanism: tie open-model subsidies to audited training compute.
Stakeholders: labs, smaller developers, users, taxpayers, the agency.
Indicator: original-source releases per unit of public spending.
Baseline (simulated): flat reward → 4% of prize reaches sources (484/source); indexed → 98% (18.6/source).
Payment (k=4): imitator cost C_d=1.44 → reward kE_d=3.94; innovator cost C_n=5.67 → reward kE_n=18.52; total public spending ≈ 354,763.
Risks: misreported compute, duplication, crowding out distillation.
Safeguards: audit, capped pilot, publish results, revise or stop.
Intended benefit, not yet demonstrated.



Future Research Roadmap

Limitation: illustrative parameters, simulated firms, manipulable compute.
Economics: which reward raises sources/$? — preregistered policy pilot.
Computer science: can original compute be verified? — provenance logs + adversarial benchmarks.
Behavioral: how do firms adapt over rounds? — repeated-choice experiments.
First task: an auditable training measure.

OCT 12 duet defense · OCT 14 roundtable · OCT 21 final

OPEN MATERIALS

Hugging Face: huggingface.co/spaces/dku-comsci-econ206-2026/FP4_PS2
GitHub: github.com/dku-comsci-econ206-Autumn2026/PS2-FP4-CapabilityTransfer
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